

Unbounded Viewstamped Replication Revisited for Diskless Strong Consistency

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Abstract—Small replicated metadata should not require a disk flush on the critical path. Unbounded Viewstamped Replication Revisited is a protocol design for embedding strong consistency with volatile normal-path replication and crash-stop reincarnation. A cleanly stopped process restarts with its identity and flushed state. A crashed process permanently retires its old identity and rejoins under a fresh identity, obtaining state and voting authority through reconfiguration. Its organising mechanism is leader overlap: where the leader is the sole intersection of the active decision quorum and the next prepare quorum, it withholds its own promise so that preparation can proceed while the active quorum remains usable. Five-voter replacement uses six committed batches, including exact doubling and halving. A constructive Rust solver also computes paths between arbitrary weighted-majority configurations with an available majority at each endpoint. The Lean development proves weighted reachability and a phantom-identity casting-vote construction, alongside agreement and identity exclusion for a simplified replacement instantiation. Appendix B separates these contracts and presents executable checks.

Index Terms—Crash-stop, leader overlap, reconfiguration, strong consistency, Viewstamped Replication.

I. INTRODUCTION

Oki and Liskov introduced Viewstamped Replication in 1988 [1]. Liskov and Cowling’s 2012 Viewstamped Replication Revisited (VRR) eliminated disk writes during normal processing and view changes [2]. Replication therefore retained the state needed for agreement without adding a local disk flush to each update. The remaining difficulty was how to bring a replica back after loss of that state.

Michael *et al.* exposed a safety failure in VRR’s recovery mechanism in 2016 [3, Sec. 4.2 and Appendix A.1]. A replica can send a view-change commitment, crash, and return in an older view having forgotten the commitment. It can then help commit an operation that a later leader loses when delayed view-change messages arrive. Their counterexample breaks linearizability, including when channels preserve message order. The authors provide a virtual stable storage construction to support diskless crash recovery. Their result makes the problem precise: restoring a replica’s data must also preserve the commitments made by its identity.

Unbounded Viewstamped Replication Revisited (uVRR) repairs this failure by making a crash final for the protocol identity. The crashed identity never resumes execution. The process that replaces it joins under a fresh identity, receives state from the cluster, and acquires voting authority through committed membership changes. Messages sent before the crash remain messages of the old identity; the new process

cannot add fresh votes to that identity’s history. A cleanly stopped process can instead restart under its existing identity because it has preserved its protocol state. Startup fencing and clean shutdown use durable writes, keeping those writes outside normal replication. Loss of volatile state therefore leads to replacement of a member: crash recovery becomes crash-stop reincarnation.

Turner supplies the construction that connects this lifecycle to continued replication [4]. Where the leader is the sole intersection of an active decision quorum and the next prepare quorum, it can withhold its own promise while collecting the others. The active quorum remains usable during that preparation; the leader completes the handover with its own local promise. We use this leader overlap to organise reincarnation. For five unit-weight voters, the runtime schedule has six committed batches, including exact doubling and halving. The simplified two-batch instantiation retained for the concrete lifecycle theorem is stated separately in Appendix A. Appendix B supplies a general configuration-path constructor and an abstract casting-vote completion using fresh identities. Turner’s agreement argument supplies the basis for composing safe adjacent configuration boundaries.

Together, these ideas retain VRR’s diskless normal path while giving failed members a route back through fresh identities. The intended setting is agreement over small amounts of metadata embedded in a distributed application [5]. Such a service can coordinate membership or advisory-lock metadata without putting synchronous persistence on every update’s critical path. Its normal replication need not wait for a disk flush or batch updates to amortise one.

Section II develops the lifecycle and leader-overlap construction. Section III gives the five-voter schedule and its Lean proof of per-slot agreement from the protocol’s history invariants, together with the replacement weights, exclusion of the evicted voter and identity nonreuse. Section IV develops the metadata applications, and the appendix sets out the mathematical hypotheses and verification evidence. The name “unbounded” follows Turner’s treatment of pipelining through reconfiguration: agreement does not depend on a fixed bound on outstanding slots.

II. METHOD

A. Processes, state and identity

The agreement model is asynchronous and non-Byzantine. Messages may be delayed, and an old message retains the identity that issued it. Each incarnation has a logical identity distinct from its physical host. The lifecycle must ensure

that an identity never produces new protocol actions after a replacement incarnation takes over.

A clean stop first quiesces protocol processing, then flushes the state required for restart, and finally records a durable stopped marker. On startup, a valid stopped marker permits the process to reload that state and enter *Restarting* with the same identity. Before sending protocol messages it durably records that it is running again. This boot fence prevents a later crash from being mistaken for another clean restart. A restarting process retains its promises and accepted state, so ordinary timeout processing is permitted by the design.

Without a valid clean-stop marker, startup enters *Joining* under a fresh identity. The old identity is crash-stopped. Cached state may reduce transfer work, but it provides no voting authority. The joiner asks the leader to evict the old incarnation and admit the new one; the leader can send missing state while coordinating the change. A joiner does not vote or initiate a view change. Its authority comes from the committed configuration, never from the fact that it shares a host with a former voter.

The implementation design uses replicated superblock markers and majority interpretation of valid marker copies. This paper abstracts that mechanism as correct clean-stop classification and durable identity fencing. The Lean result concerns the resulting lifecycle; it does not model the storage protocol itself.

B. Quorums across eras

An era determines the quorum families for a suffix of log slots. Write $\mathcal{Q}_e^I$ for its phase-one family and $\mathcal{Q}_e^{II}$ for its phase-two family. The history model requires the following within-era and adjacent-era intersections:

$$q \cap r \neq \emptyset \quad (q \in \mathcal{Q}_e^{II}, r \in \mathcal{Q}_e^I), \quad (1)$$

$$q \cap r \neq \emptyset \quad (q \in \mathcal{Q}_e^I, r \in \mathcal{Q}_{e+1}^{II}). \quad (2)$$

The direction in (2) matters: the earlier phase-one quorum must meet the later decision quorum. These conditions are part of Turner’s era-based agreement argument [4].

Our concrete schedule uses the same strict weighted-majority family for both phases. If $w_e(n)$ is a nonnegative integer weight on a finite support, a quorum q satisfies

$$2 \sum_{n \in q} w_e(n) > \sum_n w_e(n). \quad (3)$$

The weighted proof establishes the requisite intersections when successive configurations differ by at most one unit of total absolute weight. Membership records for zero-weight nodes may change in the same batch without altering (3). A crash itself does not change any weight; only committed configuration commands do.

C. The leader as the overlap

Suppose a leader ℓ uses ballot b with an active decision quorum q . To prepare ballot $b' > b$ for the next era, it chooses a prospective phase-one quorum q' with

$$q \in \mathcal{Q}_e^{II}, \quad q' \in \mathcal{Q}_{e+1}^I, \quad q \cap q' = \{\ell\}. \quad (4)$$

TABLE I
FULL FIVE-VOTER REPLACEMENT: SIX COMMITTED BATCHES. ZERO DENOTES VOTING WEIGHT, INCLUDING IDENTITIES ABSENT FROM MEMBERSHIP.

Stage	n_0	n_1	n_2	n_3	n_4	n_5
Initial	1	1	1	1	1	0
Double	2	2	2	2	2	0
Join, increment	2	2	2	2	2	1
Decrement old	1	2	2	2	2	1
Decrement, leave	0	2	2	2	2	1
Increment new	0	2	2	2	2	2
Halve	0	1	1	1	1	1

This is Turner’s casting-vote condition [4]. The leader first sends prepare requests only to $q' \setminus \{\ell\}$. Their promises do not disable an acceptor in q . The leader withholds its own promise and may continue proposing at b , provided no other event invalidates the old ballot’s guards.

After receiving those promises, the leader chooses the suffix boundary k and makes its own promise locally. Phase one is then complete without another network exchange for the leader’s response. Proposals at b' must still obey the normal rule for preserving values reported by phase one. Figure 1 shows the message order.

Decision quorums follow the era of the *slot*, not simply the era associated with a ballot. Consequently, continuing to decide later-era slots at b also requires a quorum valid for those slots. In the concrete pivot below, the active quorum is a majority in both adjoining eras.

D. Replacing one of five voters

Let n_0 be the failed incarnation, $n_1, \dots, n_4$ the other original voters, and n_5 its fresh replacement. Table I gives all six committed boundaries of the five-voter weighted replacement. A join inserts the replacement at zero weight before its first increment; a leave removes the old identity after its final decrement. The three-voter unit-weight case uses two batches. These are distinct schedules, both covered by the Rust tests.

Each boundary preserves cross-era majority intersection by a unit change or exact scaling. Figure 1 depicts the pivot of the simplified agreement instantiation retained in Appendix A; it is not a diagram of all six weighted boundaries. A failed witness for a particular pair of quorum sets does not establish failure of reconfiguration reachability. Appendix B constructs paths with an available majority and shows how fresh logical identities supply an abstract singleton leader intersection. Membership weight and receipt of a protocol message remain distinct quantities.

III. RESULTS

A. The agreement result

The final Lean theorem composes an era-based agreement theorem with the concrete replacement weights and an abstract identity lifecycle. Its contract assumes a well-founded total ballot order, a ballot-to-era map monotone in ballot order, the required ballot/slot era bound, and the history invariants

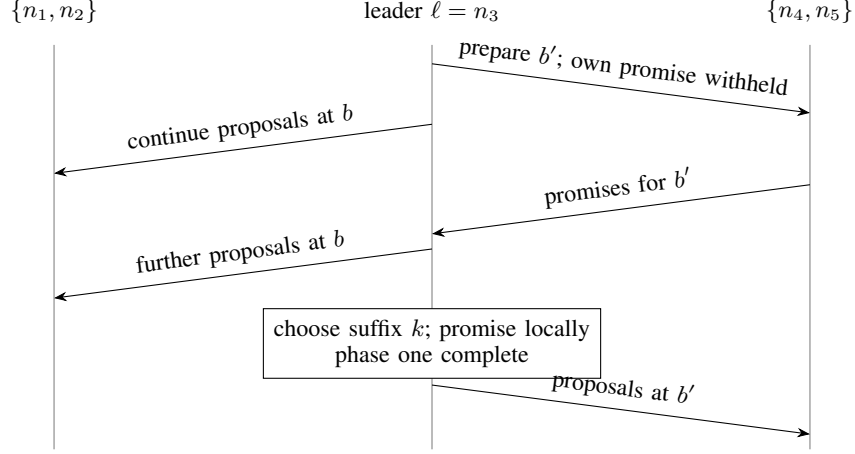


Fig. 1. Leader overlap in the simplified Appendix A schedule at $E_1 \rightarrow E_2$: $q = \{n_1, n_2, n_3\}$ and $q' = \{n_3, n_4, n_5\}$. Original schematic of Turner’s casting-vote construction [4], specialised to the replacement schedule. Time runs downwards without a scale. Group arrows represent messages to each member; acceptance replies, state transfer and unrelated traffic are omitted.

corresponding to Turner’s P2–P7 [4]. Appendix A states the assumptions and reproduction procedure. Those invariants constrain promises, reports of prior acceptances, phase-one justification and value selection, acceptance of proposals, and quorum evidence for a chosen value. The theorem derives quorum intersection from the weight schedule.

Theorem 1 (Five-voter replacement). *For a history whose phase-one and phase-two quorum families follow the simplified Table II, assuming the ballot-order, era and history conditions above, any two values chosen for the same slot are equal. In E_1 , both the failed identity and its replacement have zero weight; in E_2 , the failed identity has zero weight and its replacement has unit weight. From completion of the crossing era onwards, the old identity has no voting weight in any continuation of the forced configuration run. Separately, along an identity run satisfying the formal lifecycle transitions, an identity that has been superseded by a bump is never again the current identity.*

Proof. The strict-majority construction and one-unit boundaries establish the within-era and adjacent-era quorum intersections (1)–(2), together with quorum nonemptiness. Instantiating the era-based agreement theorem with those families establishes equality of chosen values under the supplied history invariants. Evaluation of the concrete configurations establishes the stated weights. The eviction invariant preserves zero weight for the old identity along the forced run. The identity lifecycle transitions preserve monotonic identity growth, excluding a superseded identity from becoming current again. $\square$

The leader-overlap lemmas establish that preparing members outside the active quorum preserves that quorum’s guards, and that adding the leader’s promise completes the prospective phase-one quorum.

B. Evidence and reproduction

The Lean 4 project is archived at repository commit 352b0bf [11], extended here by adding the ex-

isting eviction invariant to the final conjunction. It uses Lean 4 version 4.33.1 [12] without Mathlib. The source directory is `formal/uvrr-lean/UVRR/`. The weighted intersection, era agreement, casting-vote and simplified five-voter modules supply the components of Theorem 1. Its final statement is `ReincarnationSafety.five_safe`, in `ReincarnationSafety.lean`, importing `ReincarnationAgreement.lean`.

From `formal/uvrr-lean`, run `lake build` and `python3 check_axioms.py`. The checked baseline’s axiom audit covers 355 declarations and permits only Lean’s standard logical axioms; it rejects proof holes and custom axioms. The recorded rung-26 evidence also exercises a negative control that tries to give the replacement voting weight in E_1 . Its failure checks that particular erroneous claim. The repository contains the proof-generation transcript; generated proof terms are checked by Lean, and the generator is not part of the trusted proof checker.

Theorem 1 covers the simplified two-batch instantiation with its final configuration retained thereafter. Appendix B supplies the general weighted reachability result and the runtime solver contract. The general weighted-intersection lemmas have broader scope than this instantiation.

IV. DISCUSSION

A. Applications

The intended deployment is an embedded authority for small metadata, with ordinary replication performed in memory.

a) Short-lived advisory locks and leader leases: A replicated state machine can order acquisition, renewal and expiry decisions over a small record containing an owner, generation and expiry policy. An expiry command for an old generation must not revoke a newer grant. If a client uses ownership to control an external resource, that resource may also need to reject stale generations. A real-time leader lease additionally needs explicit clock and timing assumptions.

b) *A membership authority for another algorithm:* An application whose bulk algorithm uses gossip or another weaker consistency model may still need one agreed version of cluster membership, ownership assignments or configuration changes. An embedded replicated state machine can supply that authority, while the bulk algorithm disseminates or consumes its decisions.

c) *The CORFU precedent:* CORFU provides a concrete precedent for compact configuration metadata alongside a larger storage system [6]. Its reconfiguration is patterned after Vertical Paxos; Section 2.5 describes storage units acting as acceptors in a Paxos-like protocol. Section 2.6 estimates that adding 0.1% writes keeps projection metadata below 25 MB for a 1 TB cluster under an adversarial workload. The described implementation uses a shared network drive for agreed projections.

d) *External coordination services:* ZooKeeper and etcd are established ways to supply coordination separately from an application. ZooKeeper offers linearizable state changes and per-client ordering, while local reads have weaker semantics [7]. etcd documents strict serializability for its key-value operations and distinguishes default linearizable reads from optional serializable reads [8]. An application must choose the operation semantics it actually needs.

A separately operated service also brings deployment and resource costs. For scale, etcd’s version 3.6 hardware guide gives a small-cluster example with dedicated two-vCPU, 8 GB AWS machines [9]. etcd can itself be embedded in a Go application [10]. The proposed combination here is small application-owned state, volatile normal-path replication, and replacement through fresh identities and overlapping quorums.

B. Scope

a) *Implementation correspondence:* The final theorem assumes protocol-history invariants. Their composition with the era-changing operational model is left as an exercise to the reader, as specified in Appendix A. The appendix also identifies the corresponding implementation obligations: clean-stop classification, identity fencing and state transfer before voting.

b) *Scope and availability:* Table I replaces one unit-weight voter among five. Appendix B constructs schedules for arbitrary weighted-majority endpoints with available majorities. A higher ballot, an unavailable leader or insufficient surviving voters can interrupt progress. Fresh identities permit repeated replacement in the design; the theorem covers the displayed schedule.

c) *Meaning of strong consistency:* The mechanised result is per-slot agreement. A complete service-level linearizability argument also needs ordered execution, appropriate read semantics and a mapping between client operations and log entries. Lease guarantees require the additional conditions stated above.

V. RELATED WORK

Viewstamped Replication introduced the organisation of replication around views and a primary [1]; VRR revisits that

TABLE II
SIMPLIFIED TWO-BATCH AGREEMENT INSTANTIATION. A ZERO RECORDS VOTING WEIGHT, NOT WHETHER A ZERO-WEIGHT MEMBERSHIP RECORD IS PRESENT.

Era	n_0	n_1	n_2	n_3	n_4	n_5
E_0	1	1	1	1	1	0
E_1	0	1	1	1	1	0
E_2	0	1	1	1	1	1

organisation and separates normal operation, view change, recovery and reconfiguration [2]. Michael *et al.* identify the failure of diskless recovery to preserve view-change commitments and construct virtual stable storage [3]. uVRR takes a different route: it makes a crash final for the old identity, then brings the process back as a new member through reconfiguration. The replacement schedule connects that lifecycle to agreement across configurations.

Turner’s *Unbounded Pipelining in Dynamically Reconfigurable Paxos Clusters* supplies the central quorum geometry, era-based agreement argument and casting-vote construction [4]. Withholding the leader’s own promise is Turner’s mechanism. The present work applies it to reincarnation, proves weighted configuration reachability and an abstract casting-vote completion, and proves agreement for a concrete replacement instantiation from the protocol’s history invariants. The application discussion places this design alongside coordination services and CORFU, using their stated semantics and resource context.

APPENDIX A

PROOF STRUCTURE, HYPOTHESES AND EXECUTABLE CHECKS

This appendix separates the mathematical argument, its mechanised components, and the obligations supplied as hypotheses. The term UPaxos below refers to Turner’s paper [4]. The weighted-intersection and rescaling results follow from UPaxos Appendix A.

A. Weighted overlap, doubling and halving

Let N be a finite set of identities, with nonnegative integral weight functions w, v . Write $W = \sum_{n \in N} w(n)$, $V = \sum_{n \in N} v(n)$ and $w(q) = \sum_{n \in N \cap q} w(n)$. A strict majority has $2w(q) > W$.

Lemma 1 (Scaled overlap, UPaxos). *If positive integers a, b satisfy*

$$\sum_{n \in N} |aw(n) - bv(n)| \leq 1, \quad (5)$$

then every strict majority under w intersects every strict majority under v .

Proof. Positive scaling preserves strict majority. Suppose majorities q, r are disjoint. Summing the nodewise disjointness bound gives

$$2aw(q) + 2bv(r) \leq aW + bV + \sum_{n \in N} |aw(n) - bv(n)|. \quad (6)$$

Since all quantities are integral, the two strict majority inequalities make the left side at least $aW + bV + 2$. Equations (5) and (6) contradict that bound. The Lean declarations are `WeightedGeneral.disjoint_bound` and `WeightedGeneral.scaled_overlap`. $\square$

Taking $a = b = 1$ gives the one-unit change rule. Exact rescaling makes the distance zero: doubling all weights preserves the majority family, as does exact halving when all weights are even. Thus these transformations preserve the required intersections. Arbitrary rounding during halving is not covered by this argument. Lean provides `majority_scale` and `scaled_equal_overlap` in the same module. Appendix B applies these local closure results to arbitrary finite weighted paths; the concrete lifecycle conjunction below retains its stated two-batch instantiation.

B. Hypotheses of the agreement theorem

For each slot i and ballot b , let $e(i)$ and $e(b)$ be their eras. The contract of `ReincarnationSafety.five_safe` supplies:

- 1) A well-founded strict total ballot order, with $e(b)$ monotone in that order. A proposed value satisfies $e(b) \leq e(i)$.
- 2) Both quorum families equal the strict-majority family of the displayed E_0, E_1, E_2 sequence, with E_2 retained afterwards.
- 3) P2–P3: a free promise for a ballot excludes lower-ballot acceptances for that slot.
- 4) P4: a promise reporting a prior acceptance reports the greatest acceptance below its ballot.
- 5) P5: a proposal is justified by a phase-one quorum at its ballot era; if that quorum reports prior values, the proposal selects the value at a maximal reported ballot.
- 6) P6: every acceptance has a corresponding proposal.
- 7) P7: a chosen value has acceptance evidence from a phase-two quorum at the slot era, and $e(i) \leq e(b) + 1$.

These are history predicates. The fixed-configuration operational components in runs 13–16 do not constitute a completed composition with the changing era sequence.

a) Exercise (composition): Construct the mapping from the era-changing operational execution to this history and establish the listed predicates, including preservation of the promise fence, maximal-history selection, and installation across configuration boundaries. This composition is left as an exercise to the reader. The host must additionally implement clean-stop classification, fresh identity allocation and state transfer before voting.

C. Composition steps and conclusion

First, apply Lemma 1 to each adjacent pair in Table II, and self-overlap within each era. The case split over era 0, era 1, and later eras is `ReincarnationAgreement.sequence_p1`; it derives P1 for the sequence. Second, instantiate Turner’s Theorem 10 with P1 and the contract above to obtain per-slot agreement. Third, evaluate the four intermediate and final weights in `Replacement`. Fourth, apply `ReincarnationFive.five_evicted`

`never_voting`: after the crossing era, the old identity has zero voting authority throughout any continuation of the forced run. Finally, apply `Reincarnation.amnesia_unreachable` to exclude reuse of the pre-bump identity in the identity transition system. These are the four conjuncts now exposed by `Safe`.

Zero voting authority is a configuration property. Delayed old messages retain their old identity. The separate identity and configuration runs are abstract components; the exercise above supplies their relationship to a single executable protocol. The concrete leader-overlap witness is the E_1/E_2 pivot in Figure 1.

D. Commands and observed checks

From `formal/uvrr-lean`, the checks are:

```
lake build
python3 check_axioms.py
python3 check_mutations.py
showboat verify \
ladder/26-reincarnation-safety.md
```

The extended build completes 25 jobs. The axiom audit covers 412 declarations; the final theorem depends only on `propext`, `Classical.choice` and `Quot.sound`. The mutation suite checks rejection of an existential quorum checker, an unsafe decision family, an omitted next-slot guard, append after a view-change fence, and cross-view prepare receipt. It also checks detection of `sorryAx`, even when the compiler returns success. Rung 26 checks that promoting the replacement prematurely in E_1 is rejected.

The Leanstrat driver has two offline regressions, run from the repository root with `python3` applied to `tests/test_leanstrat_driver.py`. A mock API forces two concurrent proof loops to finish writing their candidates before either can copy its result, exposing a shared-file race deterministically. A second assertion checks that the prompt preserves project imports. Both failed on the original driver and pass with private per-invocation scratch directories and an import-preserving prompt.

E. Reported adversarial repair experiment

Table III records the externally supplied Fable 5.1 report, not a fresh experiment executed for this revision. The report used Leanstrat 1.5, a five-round budget per loop, isolated directories, and byte-identical statement checks on passing repairs. Its initial shared-directory batch was discarded because candidates crossed between jobs. The raw repair transcripts were not supplied with the report, so the table should be read with that provenance. The contrast between M1 and M2 suggests that supplying library signatures helped composition in these runs. M5 supplies a useful false-statement control.

APPENDIX B

CONSTRUCTIVE WEIGHTED RECONFIGURATION AND OPERATOR SOLVER

This second appendix gives the general construction. Its universe is a finite set of logical identities, including identities at weight zero. A configuration is a function $w : N \rightarrow \{0, 1, 2\}$. Let $L \subseteq N$ be the identities available to answer protocol

TABLE III
REPORTED ISOLATED LEANSTRAL REPAIR OUTCOMES. FAILURE MEANS
NO ACCEPTED REPAIR WITHIN THE STATED BUDGET.

Mutation and supplied context	Outcome
M1: delete final proof; file only	Fail
M2: M1 plus four lemma signatures	Pass, round 4
M3: wrong conjunction projection	Fail
M4: delete sequence P1 proof; signatures supplied	Pass, round 2; repeat failed
M5: false claim $c_1(5) = 1$	Fail, as required

messages, $A = w(L)$, and $D = w(N \setminus L)$. Availability means $A > D$. Correctness concerns intersection of quorum families; tolerance of a further host or datacentre failure is a separate property of each configuration.

A. Reachability and the operation schedule

Theorem 2 (Available-majority reachability). *For finite weight profiles s, t with the same availability set, an available strict majority at each endpoint, and a retained leader of positive weight at both endpoints, there is a finite path of single-identity increments and decrements from s to t . Every intermediate profile stays in $\{0, 1, 2\}$, retains that leader and an available strict majority, and consecutive strict-majority quorum families intersect.*

Proof. Decrease unavailable coordinates towards their targets, then increase available coordinates towards theirs. Each operation increases the available-minus-unavailable margin. Next decrease available coordinates towards their targets, then increase unavailable coordinates towards theirs. In these latter phases the margin is at least the positive target margin. Every coordinate moves monotonically between its endpoints; in particular the retained leader remains positive. Exactly $\sum_n |s(n) - t(n)|$ unit edits occur, the minimum possible using unit edits. Consecutive vectors differ in one unit, so Lemma 1 discharges cross-era intersection. Zero-weight joins and leaves preserve quorum families; exact doubling and halving provide additional shortcuts. $\square$

The Lean module `WeightedReachability.lean` proves `zero_one_two_connected` using an available leader of weight one as an intermediate profile. It also proves `UnitStep.overlap`, linking actual identity-indexed unit steps to the weighted quorum theorem. The shorter four-phase construction above is implemented in Rust and independently checked in Python; the Lean path is an existence proof, not a proof that the Rust program is the shortest-path implementation.

When membership identities or succession order differ, the Rust constructor uses the available one-voter intermediate. It removes zero-weight records, transfers authority to an available target voter if necessary, inserts the requested identities at zero weight in target order, and restores available target mass before unavailable target mass. The transfer has a two-voter stage before the former anchor loses its vote. Each membership edit is a separate boundary, keeping each endpoint union within the implementation’s 16-identity enumeration cap even

when the source and target memberships are disjoint. This cap bounds implementation cost; the mathematical result is finite-size general. The one-voter intermediate retains availability but reduces tolerance of additional failures.

B. Abstract casting votes with fresh identities

Fix an available leader of weight $l > 0$ and an available preparing quorum P of mass p , containing the leader. Let $T = A + D$. Add fresh identities outside P with total weight

$$\delta = \max(0, 2p - 2l + 1 - T). \quad (7)$$

Then P and $(N \setminus P) \cup \{\text{leader}\}$ are strict majorities in the enlarged universe and their intersection is exactly the leader. Indeed $2p > T + \delta$ and $2(T + \delta - p + l) > T + \delta$. The first inequality follows from the original majority and $l \geq 1$; the second is precisely the lower bound defining δ . Every prefix of the added mass keeps P a majority. Only identities in P are required to answer preparation; fresh abstract identities are not counted as received promises.

Choose P by starting with the leader and adding available identities until its mass first exceeds $T/2$. Since each weight is at most two, $p \leq \lfloor T/2 \rfloor + 2$, whence $\delta \leq 3$. Two fresh identities, with weights at most two each, suffice. The bound is sharp: available weights $(1, 2, 2)$ with the weight-one leader and unavailable weight one require three added units. Lean proves the prefix selection and this bound in `cap_two_phantom_construction`, the arithmetic minimality for a chosen P , and the set-intersection statement. This is an abstract quorum witness, not an assertion that absent hardware responds to messages. The ordinary prepare and acceptance history obligations still govern its protocol use.

C. Equal total mass and datacentre resilience

Equal total mass alone does not imply cross-era intersection. For $(1, 2, 1, 2) \rightarrow (2, 1, 2, 1)$ on (A, B, C, D) , $\{B, D\}$ is an old majority and $\{A, C\}$ a disjoint new majority. The solver instead returns

$$(1, 2, 1, 2) \rightarrow (2, 2, 1, 2) \rightarrow (2, 2, 2, 2) \\ \rightarrow (2, 1, 2, 2) \rightarrow (2, 1, 2, 1).$$

Each of these four boundaries is safe. For datacentres holding (A, B) and (C, D) at weights $(1, 1, 2, 2)$, loss of the second leaves mass two out of six; loss of the first leaves mass four. This asymmetry can be deliberate. Four unit voters split two per datacentre can use flexible phase-two quorums of two with phase-one quorums of three: local decisions and local leader election have different availability conditions. The solver’s guarantee is for strict weighted majorities; a flexible quorum policy must pass the core’s role-specific intersection gates separately.

D. Executable interface and validation

The dependency-free Rust module `src/solver.rs` exposes `solve` and `solve_replacement`. Results are ordered `EraStep` batches with the resulting configurations; the

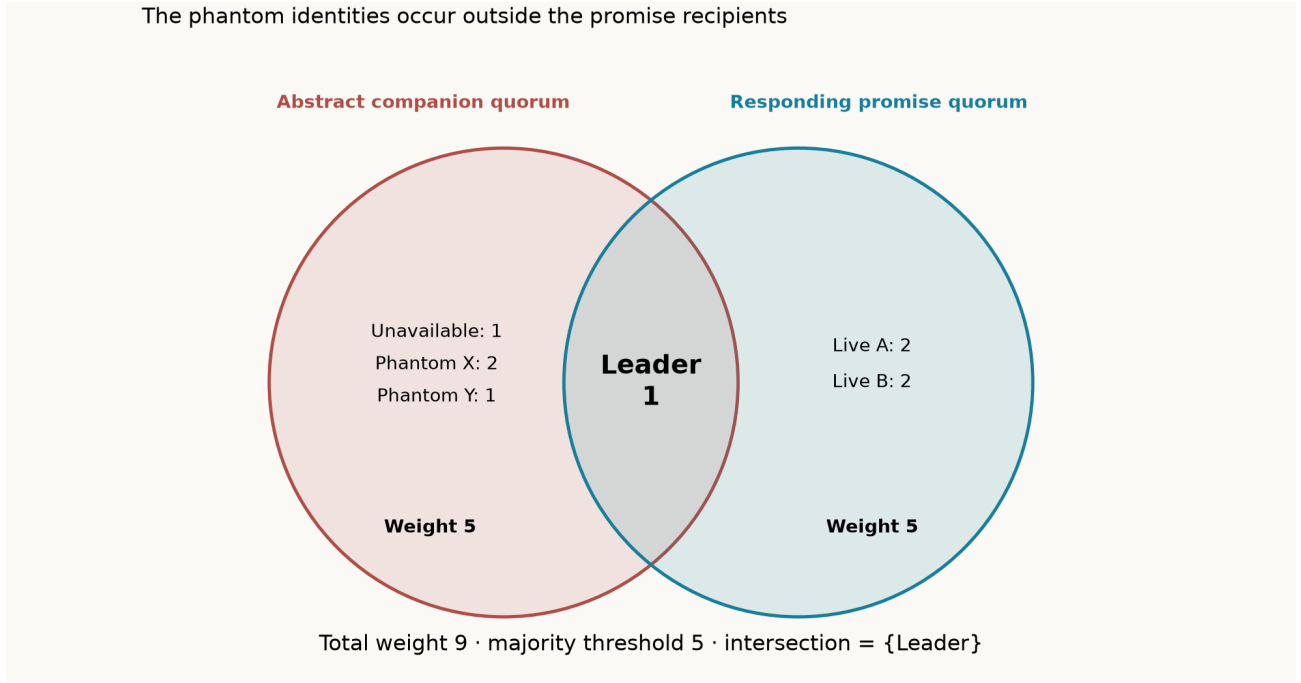


Fig. 2. Sharp phantom-mass example, generated by the Python checker. The preparing quorum uses available identities; its abstract companion includes fresh identities. Both have weight five out of nine and intersect only at the leader. Fresh identities do not supply message receipts.

target era is ignored and eras advance from the committed source. A request without an available majority at either endpoint returns the available and total masses. The replacement endpoint prefers the complete standard schedule when its intermediate configurations remain available and its endpoint matches; otherwise it invokes the general constructor. The three-unit-voter schedule has two batches and the five-unit-voter schedule has all six batches in Table I. Replanning uses the current committed configuration.

The command-line binary `reconfiguration-solver` accepts ordered `id:weight` lists, a target list or `-replace old:new`, and an available-identity list. Each output row states the batch, resulting weights and available mass. State acquisition precedes promotion. If a different leader is needed, the existing `next_view_selecting` and administrative view-change input select a later ballot directly; the ordinary quorum-backed view-change exchange still establishes leadership. A local era increment does not constitute that exchange.

Rust property tests sample arbitrary six-identity source and target weights in $\{0, 1, 2\}$ subject to available majorities, replay every batch, and check every boundary with the exhaustive quorum gate. Deterministic tests cover the equal-mass counterexample, all 36 original-leader/failed-host combinations for two nodes in each of three datacentres, exact standard schedule lengths, and disjoint 16-member endpoints. The protocol test drives all 36 unit-weight cases through leader timeout when required, reincarnation announcements, view changes and client commits. Datacentre-loss assertions describe the restored uniform configuration, not every intermediate schedule.

The independent Python checker and generated figures are in `research/weighted-reachability/`. Its

exhaustive run covers 310,007 symmetry representatives through ten identities (14,541,815 labelled profiles) and 774,159 endpoint-pair representatives through six identities (3,939,756 labelled pairs), with zero failed checks. These finite counts are computational evidence; the general claim rests on the proof above and the separately stated Lean theorems. Reproduce the Rust checks with `cargo test -test reconfiguration_solver -test reconfiguration_stepwise`, and the proof checks with `lake build` and `python3 check_axioms.py` in the Lean project. The Leanstrat driver’s check command also accepts the module without proof holes or compiled-decision shortcuts.

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